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MILITARY SPECIFICATION

ANTENNA SUBSYSTEMS FOR AIRBORNE IDENTIFICATION AND NAVIGATION EQUIPMENTS, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the general design, performance, and flight test requirements for antenna subsystems used separately and jointly with identification and navigation equipments in the frequency range of 950 to 1220 MHz.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-B-5087	-Bonding, Electrical, and lightning Protection, For Aerospace Systems
MIL-B-5400	-Electronic Equipment, Airborne, General Specification For
MIL-C-39012/16	-Connector, Coaxial, Radio Frequency (Series BNC (Cabled)-Plug Male, Class II)
MIL-C-39012/17	-Connector, Coaxial, Radio Frequency (Series BNC (Cabled)-Plug Female, Class II).

STANDARDS

MILITARY

MIL-STD-490	-Specification Practices
MIL-STD-810	-Environmental Test Methods
MIL-STD-831	-Test Reports, Preparation Of
MIL-STD-877	-Antenna Subsystems, Airborne, Criteria For Design and Location Of.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer,

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish first article inspection and approval (see 4.4.1 and 6.2.1b.)

3.2 General documents. The requirements of MIL-STD-877 apply as requirements of this specification. The requirements of MIL-E-5400 apply to antenna components containing electronic circuitry. When the general documents conflict with this specification, this specification shall govern.

3.3 Design and construction. The antenna subsystems specified herein shall be designed to be compatible with the identification and navigation requirements for the weapon system under consideration. Each antenna subsystem may be a type that is installed external to the aircraft skin or a zero drag type installed flush with the aircraft skin. Each antenna subsystem may be in physical combination with other frequency range antenna subsystems, thus serving two or more equipments, or combined together into one subsystem that will serve both the identification and navigation equipments operating in the same frequency range. If antenna subsystems are combined, it shall be possible to operate both equipment simultaneously and without interference or appreciable degradation in performance of either equipment.

3.3.1 Antenna subsystem composition. The antenna subsystems shall include all components, such as radiating elements, transmission line, matching devices, filters, and switching devices, that are connected in the subsystems beyond the output connector of the associated equipment.

3.3.1.1 Antenna radiating elements. The antenna radiating elements shall consist of slot, cavity, stub, dipole, array, or other elements suitable for installation in the aircraft weapon system consistent with the position in which it is to be installed.

3.3.1.2 Transmission line and connectors. The antenna subsystem shall be so designed and installed that connection can be readily made to the associated radio equipment through a minimum length of radio-frequency cable. R-f cable and associated connectors shall be subject to prior approval by the procuring activity. The method used in effecting electrical connection between the antenna element and the r-f cable shall satisfy the environmental requirements specified herein. The maximum allowable loss between transceiver and antenna element shall not exceed 2 dB. The use of standard cables and connectors shall in no way be construed as justification for exceeding the maximum allowable attenuation stated herein.

3.3.1.3 Selective isolating devices. The supplier shall provide selective isolating devices, as a part of the subsystem, that are necessary to assure compliance with the interference requirements specified in MIL-STD-877, and to permit identification and navigation equipments to operate simultaneously with performance equal to that obtained by operating each equipment individually.

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3.3.1.4 Structural plastic materials. Plastic materials used in the antenna design shall be in accordance with military specification where possible. However, if electrical, environmental, and structural requirements can not be met with the use of these materials, other materials not covered by military specifications may be used subject to the approval of the procuring activity.

3.3.1.5 Test point. Means shall be provided as an integral part of each antenna element for insertion of radio-frequency test signals. The decoupling between the test input and the antenna element or transmission line shall be not less than 16 dB, nor more than 20 dB. If the point of test signal input is inaccessible, an extension cable using Plug M39012/17-0001 shall be provided for easy access. The actual decoupling, including extension cable loss, shall be indicated on a permanent information plate at the point of access. The test signal connector shall be suitable for use with Plug M39012/16-0001.

3.3.2 Flush antenna designs. Flush antennas of special curvature to mate with the skin shall be designed for a given aircraft location only when standard curved shapes are not applicable.

3.3.2.1 Plastic cover plates, not an integral part of the flush antenna, shall not be used to obtain a mating surface with the aircraft skin.

3.3.3 Aerodynamics

3.3.3.1 The total aerodynamic drag, resulting from the protrusion of antenna elements into the airstream when installed on the aircraft, shall be minimized by proper shape design.

3.4 Performance

3.4.1 Electrical requirements

3.4.1.1 Frequency. The antenna rediating elements shall be designed to cover the frequency range of 950 to 1220 MHz. Other components of the antenna subsystems shall be designed to cover the frequency ranges required by their specific functions.

3.4.1.2 Polarization. When properly isntalled on an aircraft, the antenna subsystems shall transmit or receive vertically polarized signals with the transmission of horizontally polarized signals at least 14 dB below the level of the vertically polarized signals when measured as specified herein. The antennas or antenna subsystems shall be so located and installed as not to physically deviate more than 15° from the desired plane or axis of polarization when the aircraft is in normal flight attitude.

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3.4.1.3 Voltage standing wave ratio (VSWR). The design and installation of the antenna subsystems shall be such that the VSWR produced on any part of the interconnecting r-f transmission line shall not exceed 2:1 in the frequency ranges of 950 to 1000 MHz and 1100 to 1220 MHz with reference to a nominal 50-ohm level. The VSWR in the frequency range of 1000 to 1100 MHz shall not exceed 1.8:1. The requirements specified herein do not include switching transients.

3.4.1.4 Power handling capability. All components of the antenna subsystems shall be capable of handling the maximum peak and average power associated with the equipment to be installed in the weapon system under consideration and as specified in the detail specification. The power handling capability shall be demonstrated during any combination of the environmental conditions specified herein.

3.4.1.5 Radiation pattern requirements

3.4.1.5.1 Pattern coverage and gain requirements, fighter-type weapon systems. The gain requirements specified herein shall apply to fighter-type weapon systems and for any other type of aircraft that requires more complete coverage because of the operational mission it must perform. All external store configurations that are proposed or normally carried operationally shall be considered separately and each configuration shall meet the gain requirements specified herein.

3.4.1.5.1.1 Pattern measurement parameters. The radiation patterns shall be essentially omnidirectional, with the maximum null depth not to exceed 20 dB as a design goal, and with maximum radiation in the zone of $\pm 45^{o}$ in elevation from the horizon (aircraft fuselage axis at the horizon).

a. Pattern coverage

(1) Coverage off the sides of the aircraft shall be as important as the coverage in the fore and aft directions.

(2) Bottom coverage shall take procedence over top coverage when a compromise is required.

b. Radiation pattern coordinates (see figure 1)

(1) Angles of elevation θ varying between 0° and 180°, with 0° directly above the aircraft and 90° directly ahead of the aircraft, shall be used.

(2) Angles of azimuth $\emptyset$ varying between 0° and 360°, with 0° directly ahead of the aircraft, shall be used.

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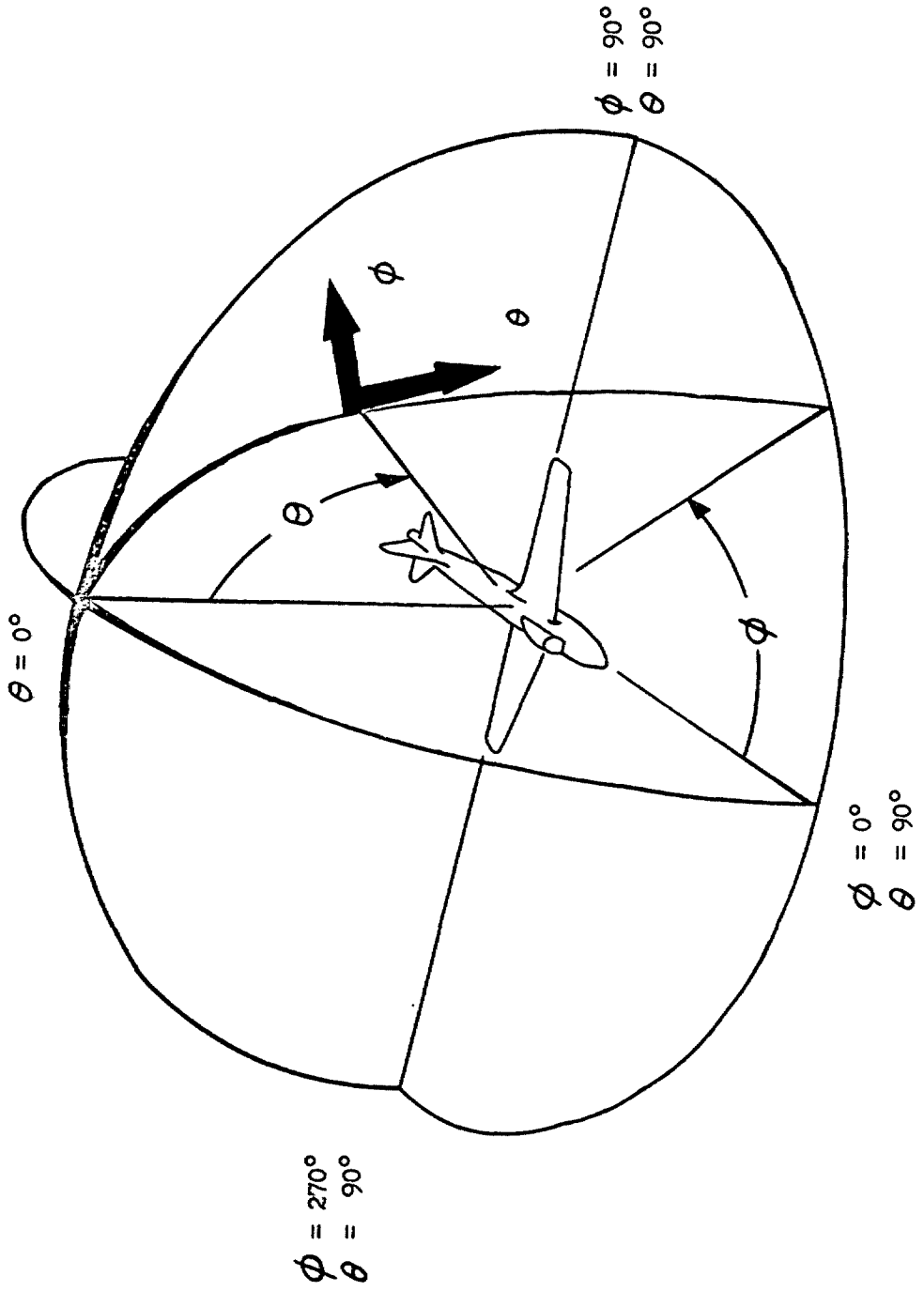


FIGURE 1. Radiation pattern coordinates

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3.4.1.5.1.2 Gain parameters

a. At any angle $\emptyset$ in the planes defined by the angle θ equals 90° to 110° with the landing gear retracted, the gain over at least 90 percent of the area shall be equal to, or greater than, the computed isotropic gain.

b. At any angle $\emptyset$ in the planes defined by the angles θ equals 110° to 135° with the landing gear retracted, the gain over at least 90 percent of the area shall be not more than 2 dB below the computed isotropic gain.

c. At any angle $\emptyset$ in the planes defined by the angles θ equals 45° to 70°, the gain over at least 90 percent of the area shall be not more than 2 dB below the computed isotropic gain.

d. With the landing gear extended, an additional gain reduction of 2 dB shall be permitted in the region defined by θ equals 90° to 120° and $\emptyset$ equals $180^{\circ} \pm 30^{\circ}$.

3.4.1.5.2 Pattern coverage and gain requirements, bomber and cargo-type aircraft. The pattern gain requirements specified herein shall apply to bomber and cargo-type aircraft.

3.4.1.5.2.1 Pattern measurement parameters. The radiation patterns shall be essentially omnidirectional, with the maximum null depth not to exceed 20 dB as a design goal, and with maximum radiation in the zone of $\pm 30^{\circ}$ in elevation from the horizon (aircraft fuselage axis at the horizon).

a. Pattern coverage

(1) Coverage fore and aft shall take precedence over coverage to the sides.

(2) Bottom coverage shall take precedence over top coverage.

b. Radiation pattern coordinates (see figure 1)

(1) Angles of elevation θ varying between 0° and 180°, with 0° directly above the aircraft and 90° directly ahead of the aircraft, shall be used.

(2) Angles of azimuth $\emptyset$ varying between 0° and 360°, with 0° directly ahead of the aircraft, shall be used.

3.4.1.5.2.2 Gain parameters

a. At any angle $\emptyset$ in the planes defined by the angles θ equals 90° to 105° with the landing gear retracted, the gain over at least 90 percent of the area shall be equal to, or greater than, the computed isotropic gain.

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b. At any angle $\emptyset$ in the planes defined by the angles θ equals 105° to 120° with the landing gear retracted, the gain over at least 90 percent of the area shall be not more than 2 dB below the computed isotropic gain.

c. At any angle $\emptyset$ in the planes defined by the angles θ equals 75° to 60° with the landing gear retracted, the gain over at least 90 percent of the area shall be not more than 2 dB below the computed isotropic gain.

d. With the landing gear extended, an additional gain reduction of 2 db shall be permitted in the region defined by θ equals 90° to 105° and $\emptyset$ equals $180^{\circ} \pm 30^{\circ}$.

3.4.2 Environmental conditions. Unless otherwise specified for a particular weapon system, the antenna subsystems, including all components thereof, shall withstand the environmental test requirements defined under 4.4.3.3 without reducing the performance below the limits specified herein. In addition to the environmental conditions specified above, the antenna subsystems shall be designed and constructed to withstand the severest environmental conditions existing in the area of the aircraft in which it is installed. Vibration, shock, extreme temperatures, effects of gunfire or other armament originating within the aircraft or resulting from flight conditions shall be taken into consideration.

3.4.2.1 Lightning protection for external antennas. External antennas shall be designed to avoid lightning currents entering the vehicle and causing damage to the vehicle or equipment installed as required by MIL-B-5087. External antennas shall be metallic in design to carry lightning currents to the air vehicle skin, or shall be designed, where possible, with shunt matching devices that will provide a path for lightning current to the air vehicle skin.

3.4.3 Flight performance. When installed on the aircraft, the antenna subsystems shall provide satisfactory navigation and IFF system performance when flight tested as specified in section 4 herein.

3.5 Size and weight. The size and weight of all components required for the antenna subsystems shall be a minimum consistent with the requirements of this specification and the detail specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspection requirements specified herein shall be classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5).

4.3 Pretesting requirements

4.3.1 NAV and IFF equipments bench and preflight tests. The navigation and IFF equipments shall be tested and adjusted within the limits of the bench and preflight procedures specified in the applicable bench, preflight, and flight test specification to be furnished by the procuring activity and the requirements specified herein. The power output and sensitivity shall be measured and recorded before each series of tests. The preflight procedures shall be conducted after the installation test of 4.3.2.

4.3.2 Aircraft installation test. The navigation and IFF equipments, including antenna subsystems, switches, selectors, and any other devices shall be properly installed in the aircraft and connected in accordance with the installation wiring diagram and installation specification to be furnished by the procuring activity. The incident powers, reflected powers, and sensitivities of both the navigation and IFF systems shall be measured prior to each flight test, using the test points specified in 3.3.1.5 and applicable test equipments to be specified by the procuring activity.

4.4 First article inspection

4.4.1 First article samples. The first article samples shall consist of models of all major components of the antenna subsystems representative of the production equipment and covered by the antenna subsystems component specification required in 4.7.1. The first article samples shall be tested under the test conditions specified herein and at the point designated by the contract.

4.4.1.1 Testing instructions. Unless otherwise specified in the contract, a minimum of three first article samples shall be required and testing shall be as follows:

a. One first article sample shall be subjected to the environmental tests specified under 4.4.3.3.

b. One first article antenna subsystem shall be installed on the test aircraft and subjected to the flight tests specified under 4.4.3.4.

c. One first article sample shall be subjected to the electrical tests specified in 4.4.3.2.

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4.4.2 Scale models. A scale model of the antennas or antenna subsystems installed on a scale model of the aircraft involved shall be used for radiation pattern measurement (see 4.4.3.1). Modeling details such as external stores, canopy constructions, flaps, wheel doors, et cetera shall be similar enough to full-scale details to permit reasonably accurate radiation patterns to be measured. The type of construction and metallic surfacing used shall permit maximum versatility for testing new locations for antennas. Reduced scale dimensions of the model shall not differ by more than 2 percent from the correct dimension, or one-sixteenth inch, whichever is greater.

4.4.3 First article tests

4.4.3.1 Radiation pattern measurement tests. Radiation patterns (see 3.4.1.5) of the antennas or antenna subsystems in the locations proposed by the contractor shall be obtained by scale model techniques. A complete integral set of patterns (voltage) shall be submitted for each antenna element used separately, and for an antenna subsystem utilizing two or more antenna elements. The isotropic gain shall be computed and plotted on each pattern chart for each separate antenna element or antenna subsystem measured.

4.4.3.1.1 When two or more antennas are used to form one antenna subsystem, the patterns of each shall be measured using the same gain setting of the pattern measuring equipment, or shall be corrected to the same gain level. The combined measured pattern of the antennas shall conform to the over-all pattern coverage requirements specified herein.

4.4.3.1.2 In combining the patterns in an antenna subsystem, the method of coupling the individual antenna elements to the equipment shall be chosen considering the effect on the resultant pattern coverage and the net gain of the antenna subsystem, including all components such as filters, isolating devices, cables, connectors, and coupling devices. The method of coupling proposed shall be subject to the approval of the procuring activity.

4.4.3.1.3 The model frequencies shall be chosen to simulate both the upper and lower frequencies and at least one intermediate frequency in the 950-to 1220-MHz band.

4.4.3.1.4 All other antennas in the immediate vicinity of the model antenna under test shall be installed and properly terminated to insure that normal coupling effects will be reflected in the pattern data.

4.4.3.1.5 Polarization test. Radiation patterns shall be taken of both polarizations in the plane defined by θ equals 90° to determine compliance with 3.4.1.2.

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4.4.3.2 Electrical tests. The procedures and application of electrical tests to determine proper performance and compliance with the electrical requirements (3.4.1) of this specification shall be prepared by the contractor and subject to approval by the procuring activity.

4.4.3.3 Environmental tests

4.4.3.3.1 The following environmental testing shall be conducted in accordance with the indicated methods of MIL-STD-810. The procedures may be modified as necessary to be compatible with any expected extreme conditions:

a. Temperature shock, method 503, procedure I

b. Temperature-altitude, method 504, procedure I, table 504-I, equipment class of applicable detail specification.

c. Rain, method 506, procedure I

d. Humidity, method 507, procedure I

e. Fungus, method 508, procedure I

f. Salt fog, method 509, procedure I

g. Dust, method 510, procedure I

h. Explosive atmosphere, method 511, procedure I for equipment not hermetically sealed; procedure II for equipment with cases designed to prevent flame and explosion propagation.

i. Vibration, method 514.1, equipment category and specific requirements of the applicable detail specification.

j. Shock, method 516.1, procedures I and II, figure 516.1-2, shock parameters (a) and (c)

k. Gunfire vibration, method 519.1, applicable procedure.

4.4.3.3.2 Immersion (leakage) tests. The equipment shall be immersed in a suitable liquid such as boiled ethylene glycol or water. The water temperature may be uncontrolled if the supply temperature is between 11°C and 20°C (51.8°F to 68°F). The absolute pressure of the air above the liquid shall then be reduced to approximately 1 inch of mercury and maintained for 1 minute, or until air bubbles substantially cease to be given off by the liquid, whichever is the longer. The absolute pressure shall then be increased to 2-1/2 inches

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of mercury. Any bubbles coming from within the equipment case shall be considered as leakage. Bubbles which are the result of entrapped air on the various exterior parts of the case shall not be considered as leaks. A helium leak detector or other means of test, equal or superior in sensitivity to the immersion test method described above, may be used upon approval by the procuring activity.

4.4.3.3.2.1 The test specified in 4.4.3.3.2 shall apply to all components of the antenna subsystems that are gasket sealed, pressurized, or fusion sealed. Components that are not sealed or pressurized shall be subjected to a rain test (4.4.3.3.1c.) and humidity test (4.4.3.3.1d.) in lieu of the immersion test.

4.4.3.4 Flight tests

4.4.3.4.1 Test setup

a. The IFF and navigation systems flight test aircraft shall be one of the first 10 assembled models of the weapon system under test.

b. The installation of the IFF and navigation systems, cables, and antenna subsystems, including switches, selectors, and any other necessary devices, shall be an exact duplicate of the installation proposed for production aircraft.

c. The extent of flight testing shall depend on the result of the model radiation pattern measurements. If the radiation patterns are inconclusive, all or part of the flight tests specified herein may be required as directed by the procuring activity.

d. The antenna subsystems shall be evaluated with all normally carried stores in place and with both landing gear and flaps or air brakes in extended and retracted positions.

e. The contractor shall take the necessary action to obtain frequency authorizations when contractor-controlled ground transmissions are required. Flight tests of the navigation antenna subsystem shall include at least one frequency in each of the upper and lower frequency bands.

4.4.3.4.2 Procedures

4.4.3.4.2.1 IFF equipment flight test procedures. The flight test procedures specified in the bench, preflight, and flight test specifications applicable to the installed IFF equipments shall apply. In addition, a range test, using the ranges specified in table I, shall be conducted using appropriate ground station facilities.

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TABLE I. Maximum range. 1/

Cruise altitude feet	Range nautical miles
5,000	60
10,000	95
15,000	120
20,000	135
25,000	155
30,000	170
35,000	183
40,000	195

1/ To attain these ranges it is necessary to choose a radial where no physical obstruction intervenes on the horizon.

4.4.3.4.2.2 TACAN equipment flight test procedures. The over-all flight test profile for TACAN equipment shall be as shown on figure 2, which includes the airport departure procedure, outbound route to maximum range, range performance, inbound flight, and final approach procedures, and shall apply to fighter, bomber, and cargo aircraft.

a. Airport departure procedure. The departure route depicted on figure 3 and as described herein shall be flown. After takeoff, a climb to altitudes and ranges from the station specified in figure 3 shall be made. At a range of 40 miles and 6000 feet altitude on outbound track, a figure-8 pattern (30-degree bank) shall be executed and then an outbound route to point D shall be continued. Continuous TACAN indication (DME and azimuth) shall be obtained from takeoff to point D which is 45 miles from the TACAN ground station.

b. Outbound route to maximum range. From point D, the aircraft shall continue a straight track and perform the mission profile climb procedure to the cruise altitude for the aircraft. After reaching cruise altitude, the aircraft shall fly to the maximum range point E for the altitude (see table I). During this portion of the flight test, the TACAN performance shall be nearly continuous. Due to the ground antenna lobing structure, one or two TACAN (DME or azimuth) search conditions may occur which are of short duration not exceeding 3 miles in range (additive miles) below 20,000 feet, 8 miles between 20,000 to 30,000 feet, and no greater than 12 miles above 30,000 feet.

c. Range performance. The aircraft shall perform a standard procedural turn after reaching the maximum range, point E, and take up an inbound heading to the ground station. The maximum range attained before search conditions occur on the outbound flight shall be greater than those given in table I for the altitudes flown. The outbound maximum range and the inbound maximum range shall not differ by more than 12 miles in the vicinity of 35,000 feet cruising altitude and 6 miles in the vicinity of 20,000 feet cruising altitude.

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8 EXECUTE A FIGURE 8 WITH 30 DEGREE BANK, AT THE POINTS AND DISTANCES GIVEN. (ALL DISTANCES FROM GROUND STATION) 4-PLACES.

A START PENETRATION LETDOWN ACCORDING TO TAKEOFF OF AIRCRAFT USING GEAR AND FLAPS AS REQUIRED.

LETDOWN PROCEDURE (SEE FIGURE 5).

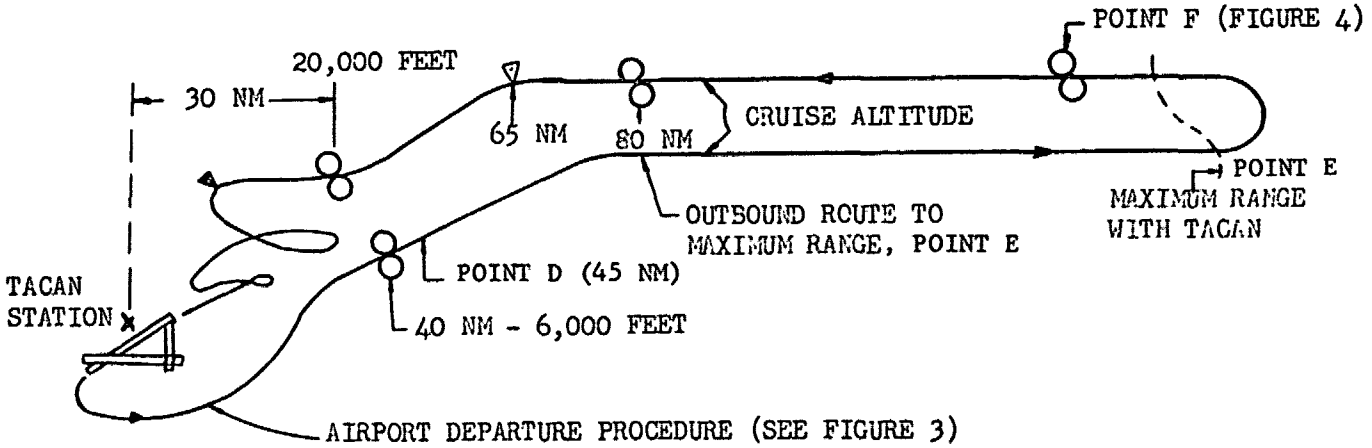
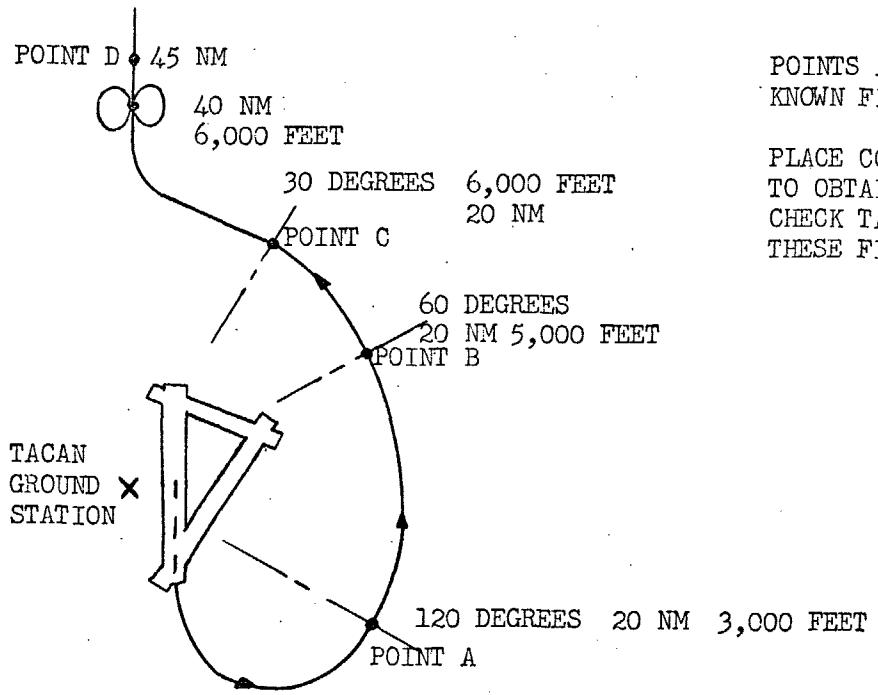


FIGURE 2. TACAN flight test plan

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POINTS A, B, AND C ARE PREDETERMINED KNOWN FIXED POINTS ON THE GROUND.

PLACE CORNER REFLECTORS AT THESE POINTS TO OBTAIN FIX USING BOMB NAV. RADAR. CHECK TACAN BEARING AND DME AGAINST THESE FIXES.

FIGURE 3. Departure plan for TACAN flight test

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d. Inbound flight. The inbound track performance shall be comparable to the outbound performance, that is, subject to the same TACAN search performances because of ground station lobing conditions. In accordance with figure 2, three figure-8 patterns shall be executed at a bank angle of 30 degrees. Continuous TACAN performance shall be maintained during the execution of the figure-8 patterns. At a distance of 65 miles from the TACAN station and at cruise altitude on the inbound radial, a jet penetration letdown to 20,000 feet shall be executed with no interruption in TACAN performance.

e. Letdown procedure. The inbound flight to the TACAN station shall be continued and letdown procedure executed in accordance with figure 4. TACAN performance shall be maintained during the entire maneuver.

4.5 Quality conformance inspections. Quality conformance inspections shall consist of individual tests and a sampling plan and tests.

4.5.1 Individual tests. Each antenna subsystem shall be subjected to the following tests:

a. Antenna subsystem component tests. The individual tests for each component used in the antenna subsystem shall be those specified in the component specifications (see 4.7.1).

b. Complete antenna subsystem tests. The individual tests for the complete antenna subsystems shall be limited to the functional testing of the IFF and navigation systems and shall be specified in the basic weapon system specification.

4.5.2 Sampling plan and tests. Unless otherwise specified in the weapon system contract, 1 production aircraft, equipped with IFF and navigation equipments, cables, and complete antenna subsystems, from each group of 50 shall be subjected to the flight tests specified under 4.4.3.4.

4.6 Rejection and retest. The procuring activity shall have the option of rejecting the antenna subsystems covered by this specification on the basis of unsatisfactory radiation patterns or flight tests results. Rejected antenna subsystems shall not be retested and resubmitted for approval without furnishing full particulars concerning the previous rejectiona and the measures taken to overcome the defects.

4.7 Data requirements. When specified in the contract, the data specified herein shall be prepared and submitted to the procuring activity for approval.

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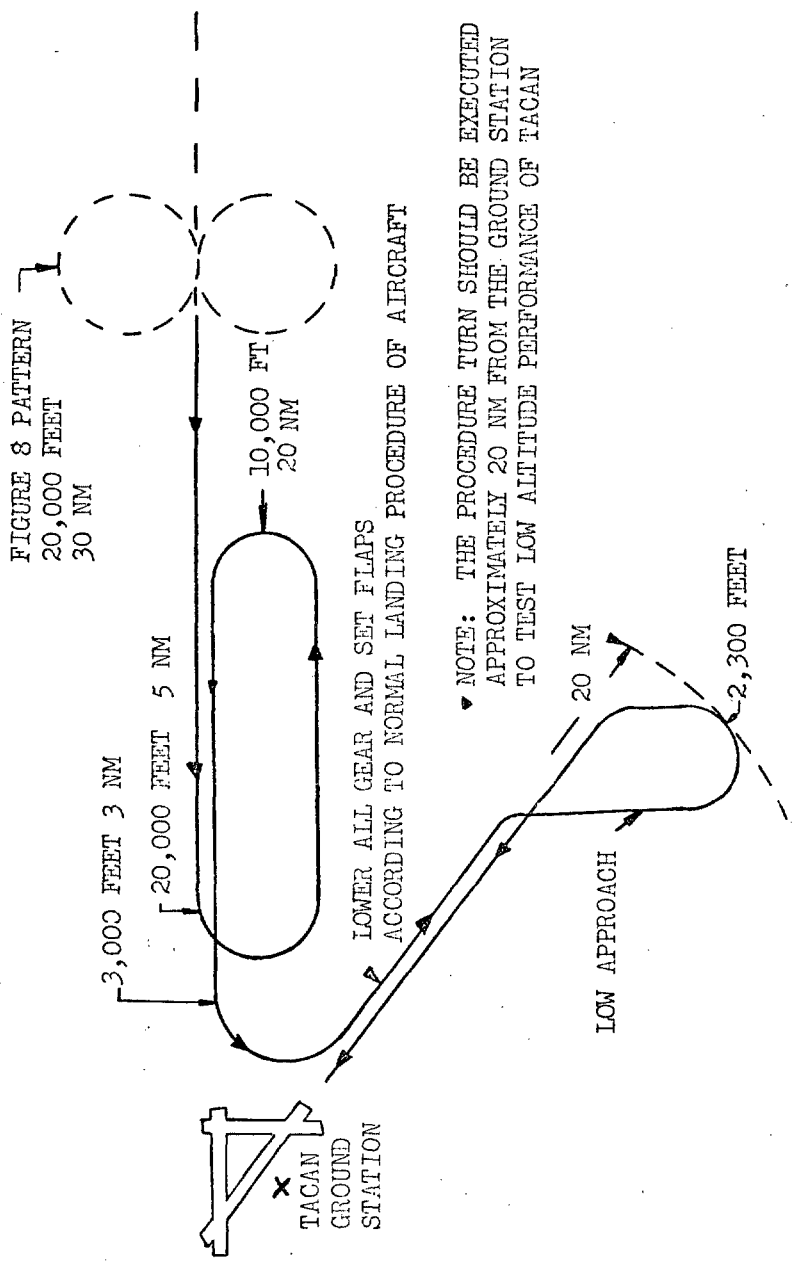


FIGURE 4. Letdown procedure.

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4.7.1 Antenna subsystem components specifications. The contractor shall prepare for each contractor-furnished equipment (CFE) nonstandard antenna or antenna subsystem component a proposed procurement specification in accordance with MIL-STD-490, outlining in detail the method for quality control. The foregoing requirement shall also apply to procurement of previously qualified standard types, including those types previously contractor-furnished aircraft equipment (CFAE). The contractor shall prepare detail specifications on all components of the antenna subsystem that are nonstandard or not previously approved; standard types not supplied from Government stock shall be included in this category.

4.7.2 Component drawings. Construction drawings, of sufficient detail to facilitate complete evaluation of any CFE antenna subsystem, including all nonstandard components, shall be submitted prior to delivery of the first production aircraft on which the antenna subsystem is mounted. The minimum acceptable drawings shall include general arrangements, outline dimensions, mounting provisions, power connections, and r-f cable lengths. Photographs shall also be considered acceptable if the object size can be closely approximated. The construction drawings may be included as part of any other report due in the same time phase.

4.7.3 Installation data. Complete information regarding the installation of the antenna subsystem in the actual aircraft shall be furnished. Information supplied shall include the type and quantity of r-f cable, connectors, relays, filters, adapters, or other miscellaneous items that form a part of the actual antenna subsystem installation.

4.7.4 Test reports. All test reports shall be prepared in accordance with MIL-STD-831.

4.7.4.1 Radiation pattern report. A detailed report covering the results of the scale model test of the recommended location shall be prepared. In addition, the contractor shall include sufficient data, both electrical and mechanical, on all other locations investigated to substantiate his choice of locations and as evidence that a thorough study has been made. These data shall be submitted as soon as possible after the first prototype structural configuration of the aircraft is completed. Additional data shall be required for each major revision of the aircraft configuration affecting the performance of the antenna subsystem.

4.7.4.2 Electrical, mechanical, and environmental test data. The contractor shall submit a detailed test report covering the electrical, mechanical, and environmental tests in accordance with the specifications required in 4.7.1.

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4.7.4.2.1 Submission of new electrical, mechanical, and environmental test data on components that have previously been approved and the application is similar shall not be required. However, the contractor shall submit previous test data and shall certify that the component(s) is capable of meeting the electrical, mechanical, and environmental performance requirements for the weapon system in which it is to be installed. Any changes affecting the electrical, mechanical, or environmental performance of the component shall render any previous approval void.

4.7.4.3 Flight testing data

4.7.4.3.1 Flight test program plan. The contractor shall prepare a detailed flight test program plan in accordance with the test requirements specified herein. This report shall include complete information on the test facilities to be used both on the ground and in the air. When more than one antenna subsystem is being tested concurrently, this program may be combined with other flight programs. The report shall be submitted prior to beginning any flight tests on the antenna subsystem.

4.7.4.3.2 Flight test report. A final test report shall be submitted analyzing and summarizing the results of all flight tests conducted by the contractor on the antenna subsystems. This report shall include the following:

a. Complete data on system losses between the transceiver and the radiating elements of the antenna subsystem.

b. Unsatisfactory results with the reasons why satisfactory results were not obtained.

c. Total number of flight hours required to perform the test. If other tests were performed during the same mission, the total hours shall be reduced accordingly.

4.7.4.4 Interim engineering reports. Iterim reports shall be prepared on the antenna subsystem development, first article inspections, quality conformance inspections, et cetera, describing in detail the progress made, including unsatisfactory results and comments on modifications required.

4.7.4.5 Final engineering reports. The contractor shall prepare final engineering reports consisting of pictorial, graphic, and written material necessary to completely describe the location, installation, and performance of CFE or Government-furnished equipment (GFE) antenna subsystems installed in the weapon systems under consideration. The contractor's identification numbers of each component of the over-all subsystems shall be included.

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5. PACKAGING

5.1 The IFF and navigation antenna subsystems shall be prepared for delivery in accordance with the instructions of the procuring activity.

6. NOTES

6.1 Intended use. The IFF and navigation antenna subsystems covered by this specification are intended for use with identification, friend or foe and TACAN equipments.

6.2 Ordering data. Procurement documents should specify the requirements herein.

6.2.1 Procurement requirements

a. Title, number, and date of this specification

b. First article samples and tests

(1) Number of first article tests

(2) Point of inspection

(3) Point of delivery for samples (samples to be submitted for approval should be shipped at least 60 days prior to production installation).

c. Selection of applicable levels of preservation and packaging, and packing.

6.2.2 Contract data requirements. When required, the data specified under 4.7 will be listed directly on a DD Form 1423 incorporated into the contract.

6.3 Typical design information. The contractor may obtain design and installation data for typical IFF and navigation antennas upon request to the procuring activity.

Custodian:
Army - EL
Navy - EC
Air Force - 11

Preparing activity:
Air Force - 11

Project No. 5985-0767

Reviewers:
Army - EL, SC
Navy - EC
Air Force - 85
DSA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

C16061

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